

SECURE CLOUD NETWORK ARCHITECTURE

Mature CISO Cheat Sheet for Exam Readiness

Cloud Models | Shared Responsibility | VPC | CASB | IaC | Zero Trust | Edge/IoT/ICS

Exam Use Notice

This PDF is built as a high-yield study sheet. The QCM section contains original Security+ style practice questions, not copied real exam questions. Learn the scenarios, not memorized dumps.

CISO Mental Model

Cloud does not remove accountability. It changes the responsibility split between the customer and the cloud provider.

Exam Focus

Expect scenario questions asking who is responsible, which control fits, and which model reduces operational burden.

One Sentence Rule

The provider secures the cloud; the customer secures what they put in the cloud.

Architecture Focus

Design for least privilege, segmentation, encryption, logging, high availability, and recovery.

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0. One-Page Revision Map

Read this page first, then use the rest for detailed memorization.

The 5-second answer pattern

When an exam question says SaaS, think identity, data, configuration, and sharing policies. When it says IaaS, think guest OS, patching, network rules, and applications. When it says PaaS, think application code, data, configuration, and platform-level access.

Concept	Remember it as	Exam trigger words	Best answer direction
Public cloud	Shared provider environment	Multi-tenant, internet, CSP	Strong IAM, segmentation, monitoring
Private cloud	Dedicated cloud for one org	Sensitive, government, bank, exclusive	More control, more cost/responsibility
Community cloud	Shared by similar organizations	Same sector, shared compliance	Shared governance and cost
IaaS	Rent infrastructure	VMs, virtual network, storage	Customer patches OS and secures apps
PaaS	Rent platform	Managed DB/app runtime	Customer secures code, data, IAM config
SaaS	Rent application	Microsoft 365, Salesforce	Customer secures users, data, sharing
CASB	Cloud security checkpoint	Shadow IT, SaaS visibility, DLP	Monitor and control cloud usage
VPC	Private network inside cloud	Subnets, route tables, endpoints	Segment workloads and reduce exposure
IaC	Infrastructure recipe	Terraform, CloudFormation, idempotent	Repeatable, auditable, consistent builds
Zero Trust	Never trust by location	Continuous verification, least privilege	Verify every session and limit blast radius

1. Cloud Deployment Models

Question to ask: who owns it, who uses it, and how isolated is it?

Model	Definition	Analogy	Security/CISO Notes
Public	Cloud services offered over the internet to multiple customers.	Apartment in a large building.	Cost-effective and scalable, but misconfiguration, weak IAM, exposed storage, and data residency must be controlled.
Hosted private	Dedicated cloud for one organization, hosted by a third party.	Private villa inside a managed resort.	Better isolation and predictable performance; still requires vendor risk management and clear contracts.
Private	Cloud infrastructure dedicated to one organization, often owned/managed by the organization.	Your own headquarters.	High control and privacy; higher cost and more internal responsibility. Common for banks/government.
Community	Cloud shared by organizations with common requirements.	Several hospitals sharing a regulated medical platform.	Useful for shared compliance needs; requires strong governance, audit rights, and responsibility clarity.
Hybrid	Combination of two or more cloud models connected together.	Company HQ plus rented cloud branch.	Common in real life. Secure interconnection, identity federation, and data flow control are critical.

Exam trap

Do not confuse public cloud with public access. Public cloud means shared CSP infrastructure; your resources can still be private if configured correctly.

2. Cloud Service Models

Question to ask: how much of the stack does the provider manage?

Model	What provider gives you	You still manage	Memory hook	Examples
IaaS	Compute, storage, networking, virtualization.	OS hardening, patching, apps, data, IAM, firewall rules, logging.	Rent the kitchen.	EC2, Azure VMs, OCI Compute
PaaS	Infrastructure plus managed runtime/database/platform .	Code, data, identity, secrets, configurations, API security.	Use a prepared pizza base.	Azure SQL, Google App Engine, managed DBs
SaaS	Complete application delivered as a service.	Users, MFA, data classification, sharing, DLP, audit review, integrations.	Order pizza from a restaurant.	M365, Salesforce, ServiceNow
FaaS/Serverless	Event-driven function execution without server management.	Function permissions, triggers, code, dependencies, secrets, logs.	Pay only when the chef cooks.	AWS Lambda, Azure Functions

CISO language

Cloud service models are not only technical choices. They decide operational burden, security ownership, audit evidence, vendor dependency, and incident response boundaries.

3. Shared Responsibility Matrix

The most exam-important idea in cloud security.

Layer / Responsibility	On-Prem	IaaS	PaaS	SaaS / FaaS
Data classification and accountability	Customer	Customer	Customer	Customer
Identity and access management	Customer	Customer	Customer	Customer / shared configuration
Application security	Customer	Customer	Customer code/config	Provider app; customer config/use
Operating system	Customer	Customer guest OS	Provider/shared	Provider
Network controls	Customer	Customer config + provider infra	Shared	Provider + customer settings
Virtualization layer	Customer	Provider	Provider	Provider
Physical servers and facility	Customer	Provider	Provider	Provider
Compliance accountability	Customer	Customer	Customer	Customer

Exam gold

If a question asks who is responsible for customer data, the answer is almost always the customer. Providers may protect the platform, but the organization remains accountable for its data and legal obligations.

4. Cloud Security Solutions

Build resilience, visibility, and controlled access into the architecture.

Topic	What it means	Use case	Security risk if wrong
High availability	Designing services to survive component failure.	Multi-AZ web app behind a load balancer.	Single point of failure causes outage.
Replication	Copying data to another location.	Backup DB replica for failover or analytics.	No recovery path after failure or corruption.
Hot storage	Fast access, higher cost.	Recent SIEM logs or active production data.	Overspending if used for old archives.
Cold storage	Slow access, lower cost.	Long-term audit archives and old backups.	Slow restore if used for urgent recovery data.
Local replication	Copy inside one data center/zone.	Protect against disk failure.	Does not protect against regional disaster.
Regional replication	Copy across multiple data centers in a region.	Improve zone/data-center resilience.	May not survive full regional outage.
Geo-redundant storage	Copy to distant region.	Disaster recovery and regional outage protection.	Cost, legal residency, and recovery testing issues.

5. VPC, Subnets, Routing, and Endpoints

Treat the cloud network like a secure building with zones and internal corridors.

Component	Meaning	Exam memory hook	Good security practice
VPC	Private logically isolated network inside a cloud provider.	Gated neighborhood inside a large city.	Separate prod/dev/security/shared services using VPCs or strong subnet segmentation.
Public subnet	Subnet with route to the internet.	Front lobby.	Place only load balancers, web front ends, or tightly controlled bastions.
Private subnet	Subnet not directly reachable from internet.	Back office.	Place databases, internal APIs, application servers, and sensitive services here.
Route table	Controls where network traffic goes.	Road signs.	Review routes to avoid unintended internet paths or lateral movement.
VPC endpoint	Private connectivity to CSP services without public internet path.	Private internal corridor.	Use for S3/storage/API access when possible to reduce public exposure.
VPC peering/transit	Connectivity between networks/VPCs.	Bridge between neighborhoods.	Use least-privilege routing and segmentation to prevent broad trust.

Architecture pattern

Internet -> public load balancer -> app tier in private subnet -> database in private subnet -> logs to SIEM. The database should not be directly exposed to the internet.

6. Cloud Firewalls and CASB

Know which control fits which scenario.

Control	Operates on	Best for	Example exam scenario
Layer 3 filtering	IP addresses and network ranges.	Basic source/destination control.	Allow only corporate VPN IPs to access admin interface.
Layer 4 filtering	TCP/UDP ports and connection state.	Port-based access control.	Allow TCP 443 but block TCP 22 from internet.
Layer 7 filtering / WAF	HTTP headers, URLs, payloads, app protocol content.	Web app/API attack blocking.	Block SQL injection or malicious request pattern.
Security group	Instance/workload-level virtual firewall.	Restrict inbound/outbound traffic close to workload.	Only app server can reach DB on required port.
Network ACL	Subnet-level stateless filtering.	Broad subnet guardrails.	Deny a bad IP range at subnet boundary.
CASB	User and cloud application activity.	SaaS visibility, DLP, shadow IT, malware scan.	Prevent upload of sensitive files to unauthorized cloud app.

CASB Mode	Where it sits	Strength	Limitation
Forward proxy	Client/network edge.	Good for managed devices and corporate networks.	Less effective for unmanaged devices outside controlled path.
Reverse proxy	Cloud application edge.	Useful for remote/unmanaged access control.	Can be complex with some apps and authentication flows.
API-based	Connects directly to cloud service APIs.	Great for scanning data, permissions, and activity after upload.	May not block inline activity in real time unless integrated with controls.

CASB shortcut

If the question says shadow IT, SaaS visibility, data exfiltration to cloud apps, risky sharing, or malware in cloud storage, think CASB.

7. Application and Data Security in Cloud

The app is still your business risk, even when the platform is managed.

- Apply secure SDLC: threat modeling, code review, dependency scanning, SAST/DAST, and secrets scanning.
- Use strong IAM: MFA, RBAC/ABAC, privileged access management, conditional access, and periodic access reviews.
- Encrypt data at rest and in transit; manage keys according to sensitivity and compliance requirements.
- Centralize logs: cloud audit logs, VPC flow logs, application logs, identity logs, and admin activity.
- Document data residency and legal implications before storing regulated data in another country.

8. SOA, Microservices, APIs, Automation, and Orchestration

Modern cloud systems are service-driven and API-driven.

Concept	Definition	Analogy	Security concern
SOA	Business functions exposed as services with defined inputs/outputs.	Company departments providing services.	Service authentication, governance, and data flow control.
Microservices	Small independent services developed, tested, and deployed separately.	Small autonomous teams.	API sprawl, secrets, service-to-service authorization, observability.
Service integration	Connecting services into complete workflows.	Departments cooperating to process a payment.	Broken trust boundaries and weak integration security.
Automation	Repeating one task reliably.	One musician playing one part.	Unsafe scripts can repeat insecure configuration at scale.
Orchestration	Coordinating multiple automated tasks.	Conductor leading an orchestra.	Misordered or overprivileged workflows can create exposure.
API	Controlled interface for systems to communicate.	Restaurant menu to request a service.	Broken auth, excessive data exposure, injection, lack of rate limits.

API Style	Format / style	Best memory hook	Security notes
SOAP	Strict XML-based messaging with formal standards.	Formal government document.	Can support mature enterprise controls; often heavier and more complex.
REST	HTTP-based operations such as GET, POST, PUT, DELETE.	Simple online form/menu.	Validate input, secure tokens, enforce authorization, rate-limit, and log.

9. Infrastructure as Code - IaC

The recipe approach to cloud infrastructure.

Term	Meaning	Exam memory hook	CISO control
IaC	Infrastructure defined and deployed through code.	Recipe for infrastructure.	Code review, version control, change approval, pipeline security.
Snowflake system	Unique manually changed system that cannot be easily reproduced.	Server with secret manual tweaks.	Eliminate by using templates, golden images, and configuration management.
Idempotence	Same call with same parameters gives same result.	Press elevator button once or five times: same outcome.	Prevents duplicate/chaotic changes and supports predictable recovery.
Immutable infrastructure	Replace systems instead of manually modifying them.	Build new clean copy, destroy old one.	Reduces drift and improves auditability.

Exam trap

IaC improves consistency, but insecure templates create insecure infrastructure repeatedly. The control is secure code review and pipeline governance, not blind automation.

10. Serverless, Fog, and Edge Computing

Where does the processing happen, and who manages the runtime?

Concept	Core idea	Good use case	Security issue
Serverless / FaaS	Run code in response to events without managing servers.	Image processing after file upload, event-driven API tasks.	Overprivileged functions, exposed triggers, weak secrets handling, dependency risk.
Fog computing	Processing node placed near IoT sensors, often between edge and cloud.	Factory node analyzes sensor alerts locally.	Distributed nodes need patching, identity, logging, and physical protection.
Edge computing	Process data close to where it is created.	Smart camera, IoT gateway, industrial router, mobile device.	Many small devices, weak patching, limited CPU, physical tampering risk.

Latency story

If an industrial sensor must stop a machine immediately, sending data to a far cloud data center may be too slow. Edge or fog processing makes the decision close to the machine.

11. Zero Trust Architecture

Never trust automatically. Always verify.

Principle	Meaning	Exam trigger words	Example
Continuous verification	Access is evaluated before and during sessions.	Always verify, dynamic risk, session context.	Require MFA when impossible travel is detected.
Least privilege	Grant only the access needed.	Need-to-know, per-session, just enough access.	Developer can access only one app environment.
Assume breach	Design as if attackers may already be inside.	No implicit trust, internal network not trusted.	Segment network and monitor east-west traffic.
Limit blast radius	Reduce damage if one account/system is compromised.	Containment, segmentation, microsegmentation.	Compromised web server cannot reach all databases.
Automation	Use context and telemetry to respond quickly.	Automated collection and response.	Risky login triggers session termination or step-up authentication.

Component	Role	Memory hook
Policy Engine - PE	Decides whether access should be granted.	The brain.
Policy Administrator - PA	Creates session-specific credential/token/instruction.	The hands.
Policy Decision Point - PDP	Decision function usually including PE and PA.	The decision office.
Policy Enforcement Point - PEP	Allows, monitors, blocks, or terminates access.	The guard at the door.
Control plane	Configures and governs infrastructure.	The command center.
Data plane	Carries actual application/user traffic.	The roads carrying traffic.

CISO warning

Zero Trust is not a product. It is an architecture: identity, device posture, policy, telemetry, segmentation, and continuous decision-making working together.

12. Embedded Systems, OT, ICS, SCADA, and IoT

Cloud exams increasingly connect architecture to physical systems and edge devices.

Term	Meaning	Security priority / risk	Example
Embedded system	Dedicated computer built for a specific function.	Limited CPU, memory, storage, battery, and patching capability.	ATM controller, smart camera, medical pump.
System on Chip - SoC	Many components integrated into one chip.	Efficient but may be hard to inspect or update.	Smartphone/IoT device chip.
FPGA	Hardware logic programmable after manufacturing.	Powerful and flexible; configuration integrity matters.	Specialized high-speed processing.
OT network	Technology that monitors/controls physical processes.	Availability and safety are critical.	Factory or water treatment network.
ICS	Industrial control systems for automated industrial workflows.	Often prioritizes availability and integrity.	Power, mining, logistics, manufacturing.
SCADA	Supervisory system for monitoring/control across sites.	Compromise can affect physical operations.	Water utility supervising pumps and valves.
IoT	Internet-connected devices with sensors/software/networking.	Weak defaults, patching issues, poor logging, supply chain risk.	Smart meter, camera, wearable.

13. ICS Security Mindset

IT protects information. OT protects operations and safety.

Environment	Typical priority order	Why it matters	Security approach
Traditional IT	Confidentiality, Integrity, Availability - CIA.	Data secrecy often drives risk decisions.	Patch aggressively when safe, protect data, monitor users.
ICS/OT	Availability, Integrity, Confidentiality - AIC.	Downtime or wrong commands can affect physical safety and operations.	Segment carefully, test patches, prioritize safety, monitor passively where needed.

Exam trap

Do not apply normal IT assumptions blindly to ICS. In OT, taking a system offline to patch can be more dangerous than delaying the patch with compensating controls.

14. Cellular IoT Quick Facts

Technology	What it is	Best use	Security notes
NB-IoT	Low-power cellular IoT based on LTE/4G concepts.	Small messages, long battery life, low bandwidth sensors.	Manage SIM identity, roaming, and device inventory.
LTE-M	Low-power cellular IoT with higher bandwidth than NB-IoT.	Tracking and mobile IoT with more frequent communication.	Monitor data use, lost devices, and unauthorized SIM/device use.
SIM identity	Subscriber identity used by cellular networks.	Identifying and authenticating cellular devices.	Treat SIMs as assets; revoke lost/unused SIMs.

15. Exam Keyword Decoder

When the exam gives you a scenario, map the wording to the concept.

Question wording	Think	Likely best answer
Employees use unsanctioned cloud apps or upload sensitive files to personal storage.	CASB, DLP, shadow IT.	Deploy CASB or enforce DLP/conditional access.
Cloud resource must access storage without public internet.	Private path to CSP service.	VPC endpoint/private endpoint.
Need repeatable cloud builds and no manual drift.	IaC and idempotence.	Terraform/CloudFormation/Bicep with version control.
Application should survive zone failure.	High availability.	Multi-AZ deployment with load balancing.
Need recoverability after regional disaster.	Geo-redundancy / DR.	Replicate to distant region and test failover.
Cloud provider handles OS and database platform, customer controls data/users.	PaaS.	Managed database service.
No direct internet access for databases.	Private subnet and segmentation.	Place DB in private subnet with restrictive rules.
Verify every session, no implicit internal trust.	Zero Trust.	Continuous verification, least privilege, PEP/PDP.
Industrial plant cannot tolerate downtime.	ICS/OT AIC priority.	Prioritize availability/safety; test controls before deployment.
Device processes sensor data locally due to latency.	Edge/fog computing.	Edge gateway or fog node.

16. CISO Design Checklist

Use this to answer architecture and audit questions.

Domain	Minimum exam-ready controls
Identity	MFA, least privilege, RBAC/ABAC, privileged access management, conditional access, access reviews.
Network	VPC segmentation, private subnets, route control, firewalls/security groups, WAF, private endpoints.
Data	Classification, encryption, key management, backup, replication, retention, DLP, legal residency review.
Monitoring	Cloud audit logs, flow logs, application logs, SIEM integration, alerting, anomaly detection.
Resilience	HA design, backups, multi-zone/multi-region architecture, tested DR, clear RTO/RPO.
DevSecOps	IaC review, pipeline controls, secret scanning, dependency scanning, container/image scanning, approval gates.
Third party	Contracts, SLA, audit rights, incident notification, data processing terms, shared responsibility matrix.
Zero Trust	Device/user verification, PEP/PDP policy flow, microsegmentation, continuous risk evaluation.

Final rule before the exam

Do not answer cloud questions only from a technology angle. Ask: who owns the risk, who operates the control, where is the data, what is exposed, and how is access verified?

17. Original Security+ Style QCM Practice

These are original scenario questions for revision, not copied real exam dumps.

- 1. A company uses Microsoft 365. The provider manages the application, but the company must control users, MFA, data sharing, and retention. Which model is this?**
 - A. IaaS
 - B. PaaS
 - C. SaaS
 - D. Community cloud
- 2. An organization deploys Linux virtual machines in a public cloud. Who is responsible for guest OS patching?**
 - A. Cloud provider only
 - B. Customer
 - C. Internet service provider
 - D. SaaS vendor
- 3. A database must not be directly reachable from the internet. Which design is best?**
 - A. Put it in a public subnet
 - B. Use a private subnet with restrictive rules
 - C. Disable logging
 - D. Use cold storage
- 4. A cloud workload must access object storage without sending traffic over the public internet. Which control fits best?**
 - A. VPC endpoint
 - B. Reverse DNS
 - C. Cold storage
 - D. FPGA
- 5. Users are uploading confidential files to unsanctioned cloud applications. Which solution provides visibility and policy enforcement?**
 - A. CASB
 - B. SCADA
 - C. FPGA
 - D. Public subnet
- 6. A firewall blocks malicious HTTP payloads and suspicious URLs. Which layer is being inspected?**
 - A. Layer 2
 - B. Layer 3
 - C. Layer 4
 - D. Layer 7
- 7. A security group allows TCP 443 to a web server and blocks TCP 22 from the internet. This is mainly what type of filtering?**
 - A. Layer 4
 - B. Physical
 - C. SCADA
 - D. Data classification
- 8. A company wants repeatable, auditable cloud deployments with no manual drift. What should it use?**
 - A. Infrastructure as Code
 - B. Shadow IT
 - C. Manual console changes
 - D. Cold storage
- 9. A manually modified server cannot be reproduced and behaves differently from other servers. What is it called?**
 - A. Snowflake system
 - B. Stateless function
 - C. CASB node
 - D. Community cloud
- 10. Running the same IaC command repeatedly produces the same desired state without duplicates. What property is this?**
 - A. Latency
 - B. Idempotence
 - C. Exfiltration
 - D. Repudiation
- 11. A function runs only when a file is uploaded and billing is based on execution time. Which model is this?**
 - A. FaaS
 - B. Private cloud
 - C. SCADA
 - D. Hosted private
- 12. A factory sensor must make a local decision instantly because cloud latency is too high. Which concept best applies?**
 - A. Cold storage
 - B. Edge computing
 - C. SaaS
 - D. SOAP
- 13. Which architecture says internal network location alone should not grant trust?**

- A. Zero Trust
- B. Flat network
- C. Local replication
- D. Public cloud

14. In Zero Trust, which component enforces allow/deny decisions at the resource boundary?

- A. PEP
- B. PE only
- C. Data lake
- D. Cold storage

15. An organization must recover from a full regional outage. Which storage approach best supports this?

- A. Local replication only
- B. Geo-redundant storage
- C. Single-zone storage
- D. Public subnet

16. A managed database service handles the database platform and OS, while the customer manages data, users, and configuration. Which model is closest?

- A. IaaS
- B. PaaS
- C. SaaS
- D. Community cloud

17. Which environment usually prioritizes availability and integrity over confidentiality?

- A. ICS/OT
- B. Marketing SaaS
- C. Personal laptop
- D. Public blog

18. A company wants to scan cloud storage for public sharing and sensitive data after files are uploaded. Which CASB mode is most suitable?

- A. API-based CASB
- B. FPGA
- C. Local replication
- D. Layer 2 switch

19. Which component is most sensitive because it configures identity, routing, security rules, and infrastructure settings?

- A. Control plane
- B. Data plane
- C. Cold archive
- D. Hot storage

20. A CISO says: cloud reduces operational burden but not legal accountability for customer data. Which concept is being emphasized?

- A. Shared responsibility
- B. No responsibility
- C. Public access
- D. Data deletion only

18. Answer Key and Last-Minute Review

Use this page in the final 10 minutes before the exam.

Q	Answer	Why
1	C	SaaS provides the application; the customer still manages identity, data, and configuration.
2	B	In IaaS, the customer normally secures and patches the guest OS.
3	B	Private subnets and restrictive security rules reduce direct exposure.
4	A	A VPC/private endpoint provides private connectivity to supported cloud services.
5	A	CASB addresses cloud app visibility, DLP, shadow IT, and SaaS control.
6	D	HTTP headers, URLs, and payloads are application-layer data.
7	A	TCP/UDP port-based decisions are Layer 4 style filtering.
8	A	IaC defines infrastructure through reviewed, versioned code.
9	A	A snowflake system is unique due to manual changes and configuration drift.
10	B	Idempotence means same input/action results in the same final state.
11	A	FaaS/serverless executes event-driven code without server management.
12	B	Edge computing processes data near where it is generated.
13	A	Zero Trust removes implicit trust and continuously verifies access.
14	A	The Policy Enforcement Point enforces access decisions.
15	B	Geo-redundant storage copies data to a distant region.
16	B	Managed database platforms are commonly PaaS.
17	A	ICS/OT often follows AIC priorities because downtime or unsafe commands can affect physical processes.
18	A	API-based CASB integrates with cloud services to inspect stored data, permissions, and activity.
19	A	The control plane governs and configures the environment.
20	A	Shared responsibility divides operational duties, but the customer remains accountable for its data and compliance obligations.

19. Ultra-Compressed Memory Hooks

Memory Hook	Meaning
Public cloud is shared, not necessarily publicly exposed.	Multi-tenant provider infrastructure can still host private resources.
IaaS = you patch the guest OS.	VMs give flexibility but more customer responsibility.
SaaS = you manage users and data.	The app is managed, but access and sharing are your risk.
PaaS = secure code, data, identity, config.	Provider handles more platform operations, not your business logic.
CASB = cloud app checkpoint.	Visibility, DLP, shadow IT, and SaaS control.
VPC endpoint = private corridor.	Access cloud services without public internet path.
IaC = secure recipe.	Repeatability is good only if the recipe is secure.
Zero Trust = airport security, not castle wall.	Verify repeatedly and limit movement after access.
ICS = safety and uptime first.	Do not blindly apply normal IT patching assumptions.

20. Reference Sources Used

Authoritative sources used to align the cheat sheet with exam and architecture terminology.

- CompTIA Security+ certification and SY0-701 exam objectives: <https://www.comptia.org/en/certifications/security/>
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- NIST SP 800-145 - The NIST Definition of Cloud Computing: <https://csrc.nist.gov/pubs/sp/800/145/final>
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